DAY 17 | PRE-READ

From My Laptop to the World

A light read before Day 17. About 15 minutes, no code required.

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Why read this?

So far, everything you built runs on your laptop or in Colab. Real users cannot reach it. This week you learn how teams ship software safely and often, and how the cloud gives you computers on demand.

1. DevOps: one team, one goal

Imagine a restaurant where the chefs cook, then hand plates to a separate team who decides weeks later whether to serve them. Food goes cold and nobody owns the result. **DevOps** joins the people who build software (development) with the people who run it (operations), so the same team cooks and serves.

2. Assembly lines for code: CI/CD

Car factories check every part automatically as it moves down the line. **Continuous Integration** does the same for code: every change is automatically built and tested. **Continuous Delivery** keeps the product always ready to ship at the press of a button.

> **Key idea:** small changes, checked automatically, shipped often, are far safer than big changes shipped rarely.

3. Renting computers: the cloud

You can own a car, lease one, take a taxi, or catch a bus. Each option gives you less to manage and less control. Cloud services work the same way:

Option	Cloud name	You look after
Own the car	On-premises	Everything, including the building and hardware
Lease a car	IaaS (Infrastructure as a Service)	The operating system, software and data
Taxi	PaaS (Platform as a Service)	Just your app and its data
Bus	SaaS (Software as a Service)	Just your account and how you use it

4. Meet Azure

Microsoft Azure is one of the largest cloud providers. You organise everything into **resource groups**, like folders, inside a **subscription**, which is how usage is billed. **Azure DevOps** is Microsoft's tool set

for planning work, storing code and running CI/CD pipelines.

5. About tomorrow's assessment

Tomorrow also includes an assessment with MCQs and coding on Weeks 2 and 3. Revisit the post-reads for Days 11 to 16 tonight.

6. Think about these before class

- What steps would you check by hand before releasing an app? Which could a machine check instead?
- Which apps that you use are SaaS?
- Why might a bank prefer IaaS, while a start-up prefers PaaS?

Kaizen (改善): continuous improvement